

①

Mid Semester Examination  
MAT 2305- Real Analysis II

①a) Suppose  $0 \leq a_n < 1$  for all  $n = 1, 2, 3, \dots$ .  
Show that if the series  $\sum_{n=1}^{\infty} a_n$  is convergent, then the series  $\sum_{n=1}^{\infty} a_n^2$  is convergent.

Let  $C_n = b_n^2 + 3b_n + 2\alpha - 1$ , where  $\alpha$  is a constant &  $b_n = 1/n^2$  for all  $n = 1, 2, 3, \dots$ . If the series  $\sum_{n=1}^{\infty} C_n$  is convergent, determine the values of  $\alpha$ .

Answer:

To prove the first statement, let's use the comparison test,

$$\begin{aligned} \text{Since } 0 \leq a_n < 1 \quad & \text{for } n = 1, 2, 3, \dots \\ \Rightarrow 0 \leq a_n^2 < a_n \quad & \text{for } n = 1, 2, 3, \dots \end{aligned}$$

Since  $\sum_{n=1}^{\infty} a_n$  is convergent &  $a_n^2 < a_n$

the series  $\sum_{n=1}^{\infty} a_n^2$  is also convergent by the Comparison test.

Second part:

We need to determine the values of  $\alpha$  for which the series  $\sum_{n=1}^{\infty} C_n$  converges.

$$C_n = b_n^2 + 3b_n + 2\alpha - 1$$

Given that;

$$b_n = \frac{1}{n^2}$$

$$\Rightarrow C_n = \left(\frac{1}{n^2}\right)^2 + 3\left(\frac{1}{n^2}\right) + 2\alpha - 1$$

$$C_n = \frac{1}{n^4} + \frac{3}{n^2} + 2\alpha - 1$$

Since  $\sum_{n=1}^{\infty} \frac{1}{n^4}$  &  $\sum_{n=1}^{\infty} \frac{1}{n^2}$  are both convergent

by P-series test with  $p > 1$ .

$\Rightarrow \sum_{n=1}^{\infty} C_n$  will converge if and only if the series

$\sum_{n=1}^{\infty} (2\alpha - 1)$  converges.

• If  $\sum_{n=1}^{\infty} C_n$  is convergent then  $\lim_{n \rightarrow \infty} C_n = 0$

$$\Rightarrow \lim_{n \rightarrow \infty} C_n = 0$$

$$\lim_{n \rightarrow \infty} (2\alpha - 1) = 0$$

$$(2\alpha - 1) = 0$$

$$2\alpha = 1$$

$$\alpha = \frac{1}{2}$$

So, the value of  $\alpha$  for which the series  $\sum_{n=1}^{\infty} C_n$  converges is  $\alpha = \frac{1}{2}$ .



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① b) Examine the convergence of the series  $\sum_{n=1}^{\infty} \frac{(-1)^n (n^2+3)}{(n^6+5n^3+7)}$

Answer:

Let's take that;

$$C_n = \frac{(n^2+3)}{(n^6+5n^3+7)} \quad \text{for } n \in \mathbb{N}$$

$$C_n > 0 \quad \text{for all } n \geq 1$$

Here,  $C_n$  is a decreasing sequence.

$$C_n > C_{n+1}$$

$$\begin{aligned} f(x) &= \frac{(x^2+3)}{(x^6+5x^3+7)} \\ f'(x) &= \frac{-4x^7 - 5x^4 - 4x}{(x^6+5x^3+7)^2} \\ f'(x) &< 0 \quad \text{for } x \geq 1 \\ \Rightarrow f(x) &\text{ is decreasing function.} \end{aligned}$$

$$\begin{aligned} \lim_{n \rightarrow \infty} C_n &= \lim_{n \rightarrow \infty} \frac{(n^2+3)}{(n^6+5n^3+7)} ; n \geq 1 \\ &= \lim_{n \rightarrow \infty} \left( \frac{1/n^4 + 3/n^6}{1 + 5/n^3 + 7/n^6} \right) \\ &= \left( \frac{0 + 0}{1 + 0 + 0} \right) \\ &= \frac{0}{1} \end{aligned}$$

$$\lim_{n \rightarrow \infty} C_n = 0$$

$\Rightarrow \sum_{n=1}^{\infty} (-1)^n C_n$  converges by Alternative Series Test.

$\Rightarrow \sum_{n=1}^{\infty} (-1)^n \frac{(n^2+3)}{(n^6+5n^3+7)}$  converges by Alternative Series Test.

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① c) Using the limit form of ratio test, examine the convergence of the series  $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} 5^n}{n!}$ .

Answer:

$$\text{Let } a_n = \frac{(-1)^{n-1} 5^n}{n!} \text{ for all } n \geq 1$$

$$a_{n+1} = \frac{(-1)^n 5^{n+1}}{(n+1)!}$$

$$\begin{aligned} \left| \frac{a_{n+1}}{a_n} \right| &= \left| \frac{(-1)^n 5^{n+1}}{(n+1)!} \times \frac{n!}{(-1)^{n-1} 5^n} \right| \\ &= \left| \frac{5 \cdot 5^n \times n!}{(n+1) \cdot n! \times 5^n} \right| = \left| \frac{5}{n+1} \right| \end{aligned}$$

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{5}{n+1}$$

$$l = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

$$= \lim_{n \rightarrow \infty} \left( \frac{5}{n+1} \right)$$

$$= \lim_{n \rightarrow \infty} \left( \frac{5/n}{1 + 1/n} \right)$$

$$= \frac{0}{1+0}$$

$$l = 0 < 1$$

$\Rightarrow \sum_{n=1}^{\infty} \left[ \frac{(-1)^{n-1} 5^n}{n!} \right]$  converges by limit form of ratio test.

① d) Use the root test to show that the series  $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{-n^2}$  is convergent.

Hint:  $\lim_{n \rightarrow \infty} \left(1 + \frac{r}{n}\right)^n = e^r$

Answer:

Let's take that

$$a_n = \left(1 + \frac{1}{n}\right)^{-n^2} \quad \text{for all } n \in \mathbb{N}$$

$$\begin{aligned} |a_n| &= \left| \left(1 + \frac{1}{n}\right)^{-n^2} \right| \\ &= \left| \frac{1}{\left(1 + \frac{1}{n}\right)^{n^2}} \right| \end{aligned}$$

$$\begin{aligned} |a_n|^{\frac{1}{n}} &= \left| \frac{1}{\left(1 + \frac{1}{n}\right)^{n^2}} \right|^{\frac{1}{n}} \\ &= \frac{1}{\left(1 + \frac{1}{n}\right)^n} \end{aligned}$$

$$\begin{aligned} \alpha &= \limsup |a_n|^{\frac{1}{n}} \\ &= \lim_{n \rightarrow \infty} \left[ \frac{1}{\left(1 + \frac{1}{n}\right)^n} \right] \end{aligned}$$

$$= \frac{1}{\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n}$$

$$= \frac{1}{e^1}$$

$$= \frac{1}{e} \quad ; \quad e \approx 2.7$$

$$\alpha = 0.3704 < 1$$

$$\alpha < 1$$

$\therefore \sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{-n^2}$  is convergent by the root test.